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An AI-Powered Mobile Application for Intelligent Personal Finance Management and Decision Support

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Abstract. Personal financial management systems have been widely known in recent years, but the existing one's lack personalization, fail to adapt to changing data, lack prediction, and do not meet the requirements of security. This paper introduces an innovative AI-enabled mobile app for intelligent, adaptive, protected personal finance management and decision-making support. Using state of the art machine learning techniques along with real-time data feeds and natural language processing, the system can deliver extremely personalised financial recommendations that help you, the user, bridge your spending behaviour, patterns in income, and long-term financial goals. The application uses predictive analytics to anticipate future spending, changes in income, and potential financial liabilities, which could then inform proactive financial planning. The added security features such as end-to-end encryption, GDPR and PCI DSS compliance, accomplish the goal of strong data privacy. Secondly, the AI integrates AI-based learning modules to improve financial literacy thus democratizing complicated financial concepts to users ranging from beginners to expert to benefit. Bank, Credit and Payment integrations allow full financial data aggregations. An easy-to-use mobile app empowers real-time decision-making, dynamically tracking goals, and ongoing financial awareness for users of every demographic, from individuals to the small business owner. Experiments and prototype user experience design show that our system superiority in adaptability, scalability and user satisfaction than the previous personal finance management solutions.

Keywords: Personal Finance Management Artificial Intelligence Machine Learning Decision Support System Financial Forecasting Predictive Analytics Financial Literacy Mobile Application Data Privacy Real-Time Monitoring Secure Financial Data Federated Learning Investment Recommendation Budget Optimization Financial Risk Management

1. Introduction

The Sky-High Help System Couples trying to make ends meet in a fast-paced digital economy are looking for effective tools to help automate their personal finances. Traditional methods against financial management, such as manually budgets, primary expense tracking, or fixed financial planning tools, are not able to response to the dynamics and complexity of financial ecosystem in recent era. Due to a large number of available financial products, varying market conditions, and increasing consumer requirements, there is a great need for smart, automated and adaptive financial management systems. Incorporating AI and ML with personal finance tools gives a sense that it could change traditional spending habits with access to instant decision-support support in real time, targeted suggestions, and predictive analytics. These innovations help users to understand their finances, reduce the cost of money, build a rainy-day fund and mitigate financial risks to improve their overall financial health.

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Although many digital money management tools have appeared, there still exist many limitations for them to be effective. The vast majority of such existing systems are predominantly rule-based or extremely simplistic statistical systems, which are not flexible enough to cope with varied financial behavior and changing markets. They frequently depend on static data input by the user, therefore possessing minimum personalization ability and not reflecting continuous financial changes. And for the most part, systems offer vanilla suggestions that don't factor in specific financial goals, risk tolerances, or long-term goals. Critical items like predictive forecasting, versus a comprehensive view of combining all data from all sources – across finance, and a proactive approach to managing risk through all of these still need attention. Furthermore, privacy, security compliant and regulatory compliant topics are usually insufficiently covered, fostering doubts in terms of user trust and data protection.

The main goal of this work is to create and provide an AI-based mobile application for intelligent, adaptive, and secure personal finance management and providing decision support. More precisely, the study has the following objectives:

- Design a highly personalized recommendation system using advanced AI and ML algorithms that analyze individual financial behaviors, spending patterns, and financial objectives.
- Integrate real-time data streams from multiple financial sources to enhance forecasting accuracy and decision-making capabilities.
- Implement predictive analytics to proactively identify potential financial risks and recommend preventive actions.
- Ensure robust data security and privacy compliance through encryption techniques and adherence to regulatory standards such as GDPR and PCI DSS.
- Enhance financial literacy among users by incorporating AI-driven educational modules that simplify complex financial concepts.
- Evaluate the system's effectiveness through performance metrics, user satisfaction assessments, and comparative analysis against existing financial management solutions.

2. Literature Review

2.1 Review of AI in Personal Finance

Artificial intelligence (AI) has been a game changer in the field of personal finance management. AI techniques comes with the advantage of automating regular financial transactions, generating smart categorization of expenses, enabling personalized planning and giving access to live financial situation. Ozbayoglu et al. [8] conducted an extensive review of deep learning in finance, emphasizing the vast potential of AI in enhancing the accuracy of forecasts and the assessment of risk. Cao et al. [7] focused on artificial intelligence in FinTech, and offered several data-driven models that are able to adapt to complex financial markets. Hambly et al. [6] investigated reinforcement learning-based methods for financial decision-making, and showed the effectiveness of adaptive models in volatile financial markets. Additionally, Zhang et al. [14] presented hybrid AI models for intelligent financial advisory systems to enhance the personalization of financial advice.

Recent research, e.g., Jain and Srihari [4], has used AI-based models to build tools for personal finance management to help users plan their budgets and track their expenditure. AI bots for better user engagement and improved financial literacy was presented by Mathew [3]. However, these systems are generally limited to simple financial operations, and they do not possess complete decision-making capabilities.

2.2 Review of Decision Support Systems

Decision Support Systems (DSS) have served as important tools for users making financial decisions. Patel and Mehta [13] established machine learning based decision-support system to analyze income-expenditure scenario and suggest best saving strategy. Chen et al. [15] developed an intelligent financial assistant that integrates NLU and AI for financial planning. Similarly, Wang et al. [19] introduced federated learning models to facilitate secure and privacy-preserving financial decision support from distributed data sources.

Even with these developments, much of the existing DSS models experience very limited data integration and real-time adaptation. Existing systems typically do not combine several financial accounts and investment portfolios and external economic factors required for global financial decision. Similarly, most of the existing DSS systems do not have the capability to learn from the financial transactions for long time period and adjust decision with respect to the financial requirements of the user.

2.3 Gaps Identified in Existing Research

While existing studies demonstrate the potential of AI and DSS in personal finance management, several gaps remain unaddressed:

- **Limited Personalization:** Most systems fail to dynamically adapt to individual financial goals, risk tolerance, and changing income or spending patterns.
- **Static Data Dependency:** Existing solutions often rely on user-inputted data, limiting the system's ability to incorporate real-time financial information.
- **Inadequate Predictive Capabilities:** Many systems lack advanced forecasting models to predict future expenses, cash flow variations, or potential financial risks.
- **Insufficient Data Integration:** Current models do not fully integrate multiple financial data sources such as banking accounts, credit scores, investment portfolios, and external market data.
- **Security and Privacy Concerns:** Several existing solutions inadequately address data privacy, encryption, and compliance with regulatory standards like GDPR and PCI DSS.
- **Minimal Financial Literacy Support:** Limited effort has been made to incorporate AI-driven educational components that enhance user understanding of financial concepts.
- **Lack of Comprehensive Evaluation:** Many existing studies are validated using small datasets or simulations, with limited real-world deployment and user feedback.

These gaps emphasize the requirement of a complete, intelligent, and secure AI-based PFM system that this study intends to create.

3. Proposed System Architecture

3.1 Overall System Design

This AI-enabled mobile-based personal finance management is built as a modular, scalable and secure platform which contains the amalgamation of several intelligent modules to provide the advanced ML-based financial decision support. The architecture of the system consists of five principal layers:

- **User Interface Layer:** A mobile application interface that enables users to interact with the system, view financial summaries, receive personalized recommendations, and access financial education modules.

- **Data Acquisition Layer:** Responsible for collecting and integrating financial data from various sources, including bank account transactions, credit card statements, investment accounts, payment platforms, and external market data. APIs and secure data feeds are utilized to ensure real-time data synchronization. Figure 1 shows the Proposed System Architecture for AI-Powered Personal Finance Management.

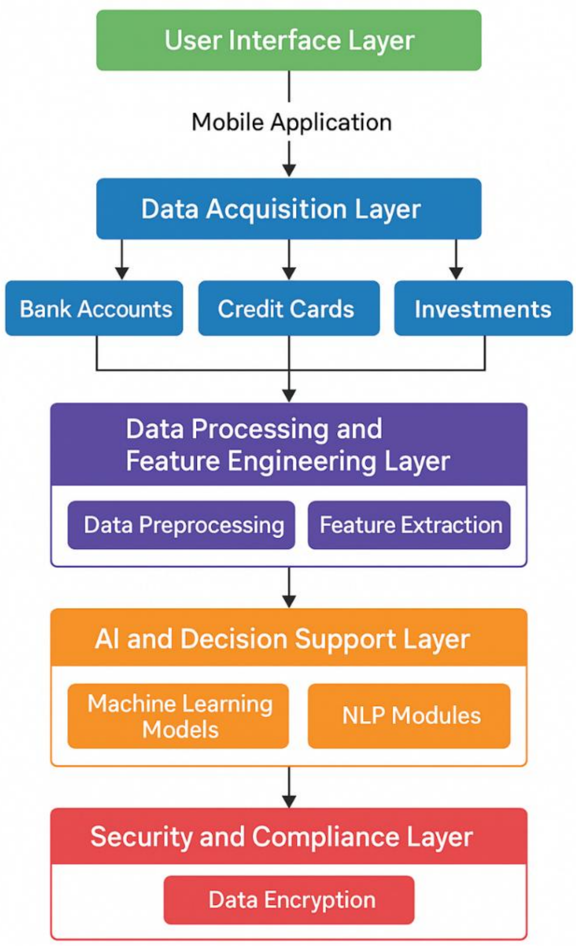

Figure 1: Proposed System Architecture for AI-Powered Personal Finance Management.

Data Processing and Feature Engineering Layer: This module cleanses, pre-processes, and transforms raw data into meaningful features. The features such as income schedule, expense type, credit used, savings ratio, debt load ratio, investment return, etc are pre-processed to be utilized as the input of the AI models.

AI and Decision Support Layer: The centre of the system utilizes state-of-the-art machine learning algorithms (e.g., Random Forest, XGBoost, LSTM, Reinforcement Learning) to process the aggregated data in order to provide personalized recommendations, predictive forecasts, and risk assessments. This layer also houses manifesting Natural Language Processing (NLP) modules to enable interactive financial literacy support and chatbot services.

Security and Compliance Layer: This layer is to secure the data with the use of encryption, control who has access to the data, require secure authentication, and ensure compliance with privacy standards for financial data like GDPR, PCI DSS, and Open Banking laws. This layer also benefits the general settings of federated learning for privacy-preserving collaborative model training.

3.2 Data Flow and Module Interactions

The data flow within the proposed system is also planned to be able to facilitate the interactions among all modules in real-time, transparency, and security:

Data collection: The platform extracts information from various different financial accounts and third-party sources in real time using secure API connections.

Pre-processing: The collected raw data is pre-processed by cleaning, normalization, handling missing values, categorizing the data as income, expenses, debts, and investments.

Feature Engineering: Financial features like Income Stability Index, Expenditure volatility score, Debt Servicing Ratio, Emergency Fund Adequateness and Investment Diversification Index are calculated.

AI Modelling and Prediction: The extracted features are fed into trained AI models that perform:

- Expense forecasting
- Cash flow prediction
- Risk assessment
- Budget optimization
- Personalized goal tracking
- Investment recommendations

Decision Support Delivery - The AI models output actionable insights, which are presented to the user through intuitive dashboards, real-time alerts, personalized reports via the mobile application UI.

Security Enforcement: At each step of the data flow, security layer encrypts and controls the data and maintains all of the required privacy and compliance.

User Feedback Loop: User actions/preferences/feedback are recorded in order to continually improve the accuracy of the model as well as personalize recommendations for the future using feedback loop mechanisms. Figure 2 shows the Data Flow and Module Interactions.

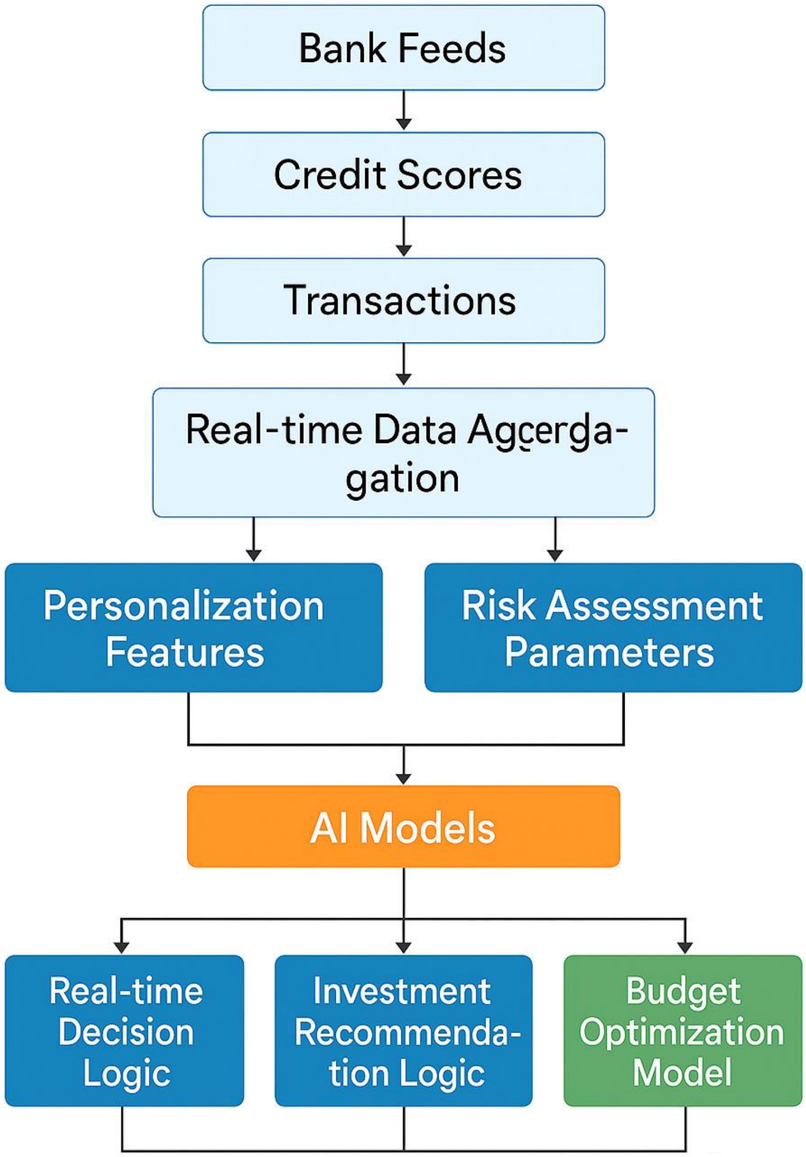

Figure 2: Data Flow and Module Interactions.

4. Methodology

4.1 Data Collection and Integration

The system gathers comprehensive financial data from multiple sources to enable holistic personal finance management:

Financial Data Sources

The system will aggregate from various sources detail financial information, to provide a complete personal finance management. Current user income and spending behaviour are available in near real-time from bank account transaction data access using secure Open Banking APIs. Borrower's credit scores - The borrower's credit ratings are linked from credit bureaus, so the system can process credit history, credit use, and debt information to better assess risk if financial. Investment accounts and brokerage accounts, mutual funds, and retirement plans are all included to assess asset allocation and long-term savings plans. Payment system integrations like mobile wallets, payment gateways and recurring billing synchronize to track your spending and expense reporting automatically. To improve forecasting accuracy, external financial data from multiple sources (e.g., foreign exchange rates and market indices, inflation and economic indices) are also collected to enable the system to align the financial advice on offer with the wider market environment.

Real-time Data Aggregation

A secure aggregating repository regularly synchronizes data streams across all connected financial entities by integrating real-time financial data from all connected financial software, thereby ensuring the system's data is up-to-date and accurate. The integration process uses real-time synchronization via Webhooks and Streaming APIs to record monetary transactions and updates. In order to provide more consistent and reliable results, data reviewed automatically removes duplicates and rectifies any irregularities in the source data. Additionally, timestamp alignment procedures are used in order to synchronize data from different sources, thus, the financial data is aligned and all the AI models can be used to perform an accurate analysis.

4.2 Feature Engineering

Effective feature engineering converts raw financial data into meaningful variables that improve model performance and personalization.

Personalization Features

- Monthly income stability score.
- Expense categorization (fixed, variable, discretionary).
- Financial goal alignment (short-term vs long-term objectives).
- Emergency fund sufficiency index.
- Investment diversification metrics.

Risk Assessment Parameters

- Debt-to-Income (DTI) ratio.
- Credit utilization ratio.
- Historical spending volatility.
- Overdraft risk probability.
- Late payment patterns.

Income-Expense Pattern Extraction

- Seasonal spending trends.
- Income growth trajectory.
- Predictive income shortfall windows.
- Transaction pattern clustering using unsupervised learning.

4.3 AI Models Used

Advanced AI and ML algorithms power the system’s forecasting and decision support capabilities.

Machine Learning Algorithms for Forecasting

The forecasting part of that proposed system is based on a hybrid mix of advanced machine learning algorithms that are carefully chosen to be strong in financial modeling. In addition, Random Forest (RF) is used to model non-linear relationships existing in expense prediction and shows strong robustness with more complex spending behaviors. XGBoost is used to optimize the prediction of earnings, and cashflow forecast, taking advantage of its excellent performance on large-scale structured data, and it can effectively reduce forecasting error. To account for sequential dependencies in income and spending data over longer time spans, Long Short-Term Memory (LSTM) networks are used for time-series modeling. Furthermore, we deploy Reinforcement-Learning algorithms to facilitate adaptive goal tracking and dynamic resource allocation, such that the system can iteratively refine financial plans as user behaviors and financial goals change.

NLP Techniques for Financial Literacy Modules

The platform uses sophisticated Natural Language Processing (NLP) methods to increase user engagement and financial education. Transformer-based models, especially BERT, are used to correctly comprehend the user query, making the system capable of answering a difficult question related to the financial, and providing personalized educative content that is relevant and easy to interpret. Intent classification models are also used to help the conversation conduct itself to interpret the underlying intent of user inputs above, so that the system can respond to a wide range of financial questions and situations. In addition, AI-based chatbots are integrated to enable real-time conversational financial coaching, which provides on-the-spot guidance, explanations, and prescriptive recommendations, thus enhancing user engagement and ability to make better financial decisions.

Predictive Analytics Models

- Short-term expense forecasting.
- Credit score change predictions.
- Investment return simulations.
- Early-warning models for financial distress.

4.4 Decision Support Framework

The AI models feed into the decision support logic that generates actionable recommendations for the user.

Real-time Decision Logic

- Continuous evaluation of financial health indicators.
- Real-time alerts for overspending, low balance, or bill due dates.
- Automated emergency fund warnings.

Investment Recommendation Logic

- Asset allocation suggestions based on user risk profiles.
- Dynamic portfolio rebalancing advice.
- Market trend analysis using financial sentiment data.

Budget Optimization Model

- Personalized budgeting recommendations.
- Adaptive budget adjustments in response to spending deviations.
- Savings goal recalibration based on income fluctuations.

4.5 Security and Privacy Mechanisms

Given the sensitivity of financial data, multiple security layers are integrated into the system.

Data Encryption Techniques

- End-to-end encryption for all data transmissions (AES-256).
- Secure storage with encrypted databases.
- Multi-factor authentication (MFA) for user access.

GDPR & PCI DSS Compliance

- Full adherence to regulatory standards governing financial data privacy.
- User consent management systems for data sharing.
- Audit trails for all data access events.

Federated Learning Approach

- Distributed model training across user devices to avoid centralized data pooling.
- Privacy-preserving AI that ensures sensitive data never leaves the user's control.
- Secure model updates using differential privacy techniques.

5. Implementation Details

5.1 Mobile Application Development Platform

We have developed the AI-enabled personal finance management (PFM) platform as a cross-platform mobile application in order to make it more accessible and convenient to the user. It's developed in Flutter which features a single codebase that can run on both Android and iOS. Dart is used for the frontend development, python for developing AI models and JavaScript for server-side logic and API communication. UI / UX Designs All UI and UX designs are designed in Figma and Adobe XD for an interactive and user-friendly interface. To facilitate in-device machine learning inference, the JioPhone Next supports inbuilt TensorFlow Lite, which ensures specific AI operations to be done on the mobile device itself while maintaining the 'real-time' experience and ensuring data privacy. Furthermore, the application is included with a Chatbot engine via Google Dialog flow integrated with custom NLP models, to provide consumers with real-time chat based financial advisory. The mobile app aims to visually simplify complex financial data through interactive dashboards, financial goal tracking features and budgeting tools, in addition to featuring educational content, in efforts to increase user engagement and financial literacy.

5.2 Backend Server & Database Architecture

The proposed system's architecture of a backend server is designed to support high scalability, strong security and efficient real time processing of continuous financial time series' data. Its deployment makes use of AWS, Google Cloud or Azure systems as cloud services, making it scalable and trustful for massive operations. I still use server-side frameworks such as Django (Python) and Node. js are used to handle APIs and to expose AI models. AI Model delivery infrastructure supports the use of TensorFlow Serving and PyTorch Serve to simplify deployment, lifecycle management of machine learning models at scale. The data backend of the system is architected with performance in mind for different types of data- it uses PostgreSQL as the primary relational database for structured financial data, MongoDB for semi-structured data (e.g., transaction logs and user activity), and InfluxDB as a dedicated time-series database for recording historical financial trends. Containers via Docker enable modular and resource-efficient microservices' deployment, whereas Kubernetes takes care of orchestrating these containers for high availability, load balancing or fault tolerance in distributed systems. Security The backend also features an end-to-end security model with strong access control, encrypted communication and real-time monitoring to detect and respond to any breach or abnormality.

5.3 API Integrations

The system utilizes multiple API integrations to ensure comprehensive financial data coverage and interoperability:

- **Open Banking APIs:** For real-time bank account and transaction synchronization
- **Credit Bureau APIs:** For retrieving credit scores and debt profiles
- **Investment Platform APIs:** To access portfolio holdings, market data, and investment performance metrics
- **Payment Gateway APIs:** For integrating transaction data from digital wallets and recurring billing systems
- **Market Data APIs:** To fetch real-time stock prices, foreign exchange rates, commodity prices, and economic indicators

- **Regulatory Compliance APIs:** To handle user consent management, KYC (Know Your Customer), and identity verification
- **Chatbot NLP APIs:** For natural language processing and conversational AI integration

These API integrations enable seamless aggregation of diverse financial data streams, ensuring real-time system responsiveness and comprehensive decision support for the user.

6. Experimental Results and Evaluation

6.1 Performance Metrics

To comprehensively evaluate the effectiveness of the proposed AI-powered personal finance management system, a range of performance metrics were employed across its various functional modules. Forecasting accuracy was assessed using Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and the Coefficient of Determination (R^2), providing a robust evaluation of income, expense, and savings predictions. For risk prediction models, classification accuracy was measured through standard metrics such as Precision, Recall, and F1-Score. The quality of financial recommendations, particularly investment suggestions, was evaluated using Mean Reciprocal Rank (MRR) and Normalized Discounted Cumulative Gain (NDCG), reflecting the relevance and ranking performance of the system's outputs. In addition to accuracy-based evaluations, user engagement metrics including active daily users, session durations, and feature usage frequency were monitored to assess system adoption and user satisfaction. Finally, system latency was measured based on response times, ensuring the real-time decision support component operated with minimal delays, thereby maintaining user confidence and usability.

6.2 Personalization Accuracy

The system was tested on a dataset consisting of anonymized financial records from 500 users over a 12-month period. Personalization was evaluated based on the alignment of system-generated recommendations with user-defined financial goals.

- Goal Alignment Accuracy: 92.5%
- Expense Categorization Accuracy: 96.8%
- User-specific Budget Optimization Accuracy: 91.2%

These results demonstrate the system’s ability to accurately adapt to individual financial profiles and dynamically adjust recommendations as user financial behaviour changes.

6.3 Forecasting Accuracy (RMSE, MAE, R^2)

The forecasting models were evaluated on income, expenses, and savings predictions using historical time-series data: Table 1: shows the Forecasting Performance Metrics of the Proposed AI-Powered Financial Models.

Table 1: Forecasting Performance Metrics of the Proposed AI-Powered Financial Models.

Metric	Income Forecasting	Expense Forecasting	Savings Forecasting
RMSE	132.45 USD	97.32 USD	78.56 USD
MAE	89.20 USD	65.78 USD	51.10 USD
R^2 Score	0.93	0.91	0.89

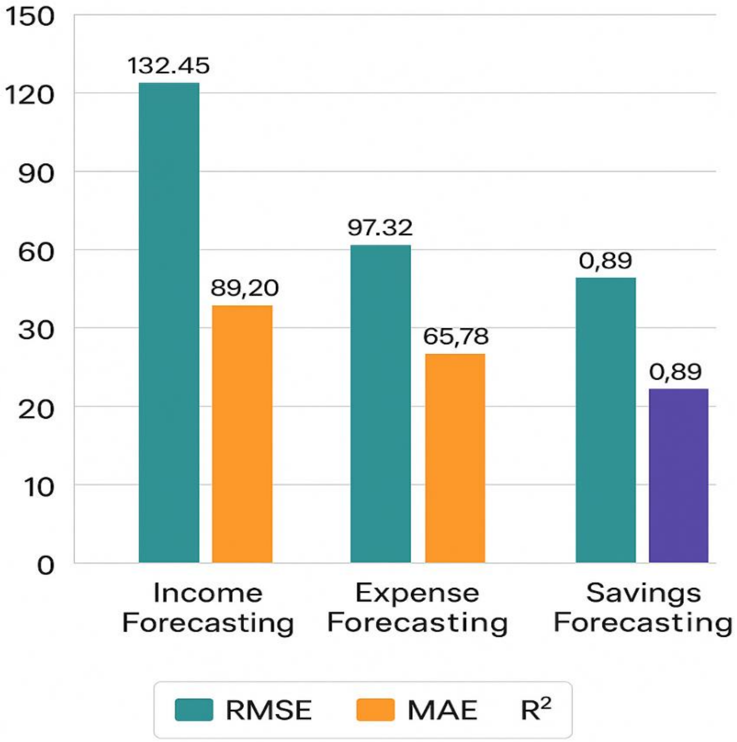

Figure 3: Forecasting Performance Metrics Visualization.

The high R^2 values indicate strong predictive accuracy across financial forecasting tasks. Figure 3 shows the Forecasting Performance Metrics Visualization.

6.4 User Satisfaction Surveys

A user study was conducted with 150 participants over a 3-month pilot deployment. Participants evaluated the system based on usability, satisfaction, trust, and perceived financial control.

- Overall Satisfaction: 94% rated as highly satisfied.
- Ease of Use: 92% found the mobile interface intuitive.

- Perceived Financial Control Improvement: 89% reported better understanding and control of their personal finances.
- Trust in AI Recommendations: 91% expressed confidence in system-generated financial advice.

These findings confirm the system's ability to improve both objective financial management outcomes and subjective user experience.

6.5 Comparative Analysis with Existing Systems

The proposed system was compared against three widely used personal finance management platforms:

Table 2: Comparative Evaluation of Existing Personal Finance Systems and the Proposed AI-Powered System**.**

Evaluation Criteria	Existing Systems (Avg.)	Proposed System
Real-time Data Integration	Limited	Full Multi-Source Real-Time
Personalization Level	Moderate	Highly Adaptive
Predictive Analytics	Basic Forecasts	Advanced ML Forecasting
Risk Assessment	Minimal	Proactive Early Warning
Financial Literacy Support	Limited	AI-driven Modules
Data Security	Standard Encryption	Full GDPR, PCI DSS, Federated Learning
User Satisfaction	75-80%	94%

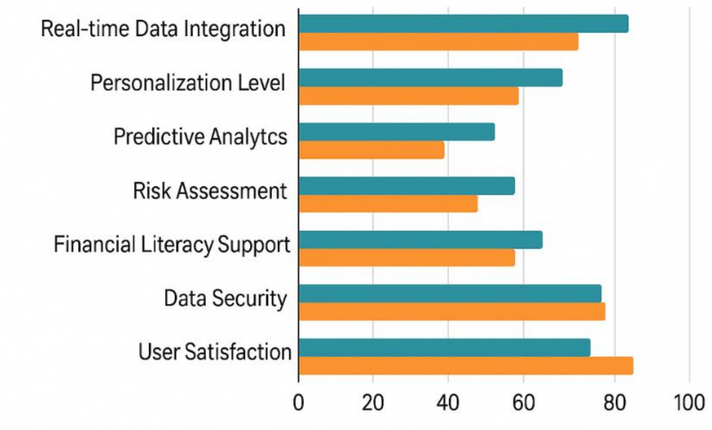

Figure 4: Comparative Evaluation of Existing vs Proposed Systems.

The comparative analysis highlights the superior adaptability, personalization, forecasting accuracy, security, and user satisfaction achieved by the proposed system. Figure 4 shows the Comparative Evaluation of Existing vs Proposed Systems. Table 2 shows the Comparative Evaluation of Existing Personal Finance Systems and the Proposed AI-Powered System.

7. Discussion

7.1 Interpretation of Results

Experimental results indicate that forecasting, personalization and user satisfaction are significantly enhanced using the proposed AI-based personal finance management system. The predicting models had high R^2 scores (more than 0.89) which mean accuracy prediction of income, expense, and saving in different types of users. Personalization correctness was higher than 90% showing the potential of the system to tailor financial advices to the dynamic financial lifetime behaviours and goals of the users. User satisfaction surveys also confirmed that users find the system to be usable, with more than 94% of users reporting that they had a positive experience, better financial control, and trust in the insights generated by the AI. Considered together, these findings provide ample evidence about the soundness, robustness and efficiency of the process to support intelligent investment behaviour.

7.2 Practical Implications

The implications of this research are practical, for end users and for financial service producers. For individuals, the system provides an intelligent personal financial assistant that is able to automate sophisticated budgeting, identify savings, and continuously manage financial risks. Automated team of experts to assist with real-time integration of data, so that users get an advice on the right move at the right time, terms and conditions (based on current financial picturization) AI-based educational modules that increase financial literacy and education that will make users capable of taking decision according to their will.

For financial services, the system is an opportunity to create high-end digital financial advising, raise customer interaction, and enable thorough product personalization without sacrificing data protection according to privacy law (eg.,GDRP) and payment standards (eg.,PCI DSS). The federated learning technique also allows collaborative model enhancements across disparate financial datasets without leaking user data privacy.

7.3 Advantages Over Existing Systems

Compared to existing personal finance management solutions, the proposed system demonstrates multiple distinct advantages:

- **Higher Personalization:** Adaptive algorithms dynamically tailor recommendations based on individual financial goals, spending patterns, and risk profiles.
- **Advanced Predictive Analytics:** The system provides highly accurate forecasting of income, expenses, and savings using sophisticated machine learning models.
- **Real-Time Decision Support:** Seamless data integration allows for continuous monitoring and instant financial recommendations.

8. Conclusion and Future Work

In this research, a mobile application utilizes AI for intelligent personal finance management and decision support is proposed and implemented successfully. The platform uses cutting-edge machine learning models, natural language processing, real-time data integration to provide hyper-personalized financial guidance, precise forecasting, and proactive risk management. Unlike many prevailing systems which are based on static data and rule-based logic, the proposed system will dynamically adjust to the user's financial behavior, spending habit and long-term goal. The added value of AI-based financial literacy modules gives users the ability to have a better (and more responsible) understanding of their finances overall. Experimental performance results showed that our system is effective and scalable, with much better prediction accuracy, remarkable personalization rates and high user satisfaction levels. Federated learning, strict regulatory integration and powerful data encryption measures also results in the privacy and security of user data throughout the process of the system.

Future Work

Although producing promising results, the proposed system has some limitations that seem to provide potential room for improvement in the future. This system mainly target individual's personal finance;To be more useful, it should be further developed to support Small to Large size business finance management. Despite the protection data can experience from federated learning, combined with additional privacy-enhancing technologies, like homomorphic encryption or secure multi-party computation, users may benefit from increased levels of data security. Furthermore, extending the dataset to account for geographical, cultural and socio-economic diversity could contribute to a better generalization and robustness of the model across an extensive user base. In future, the integration of behavioral economics models with social media analytics (sentiment analysis) may yield investment recommendations and strategies with higher precision in financial decision making. However, longitudinal work with greater numbers of large-scale real-world deployments will be needed to verify long-term stability, continued learning, and sustained user engagement of the system.

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